

Predicting the Founding of Social Organizations Among University Alumni: A Machine Learning Approach

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Abstract

Despite Mexico having one of Latin America’s largest civil society sectors, little is empirically known about what distinguishes alumni who go on to found non-profit organizations from those who do not. This gap leaves universities without actionable evidence to guide civic-engagement programs, fellowship design, or early identification of students with high social-entrepreneurial potential. We address this problem using a large-scale alumni survey ($N = 25,356$) collected by Tecnológico de Monterrey in collaboration with Quacquarelli Symonds (QS) on the institution’s 80th anniversary.

We treat nonprofit founding as a binary classification problem and evaluate six supervised learning models (Gradient Boosting, Random Forest, Logistic Regression, Decision Tree, Gaussian Naïve Bayes, and Quadratic Discriminant Analysis) after systematic feature engineering and class-imbalance handling. Gradient Boosting achieved the highest discriminative performance (ROC-AUC = 0.801, 5-fold CV). SHAP interpretability analysis identifies volunteering behavior, prior company founding, age, philanthropic donation habits, and leadership breadth as the dominant predictors. We additionally test three sociologically grounded hypotheses on leadership trajectories, multicultural education effects, and entrepreneurial archetypes using statistical tests and the trained models. Threshold sensitivity analysis shows that shifting the decision boundary from 0.5 to ≈ 0.15 recovers roughly 55% of genuine founders while maintaining precision well above the base rate of 6.95%. These findings offer a practical, data-driven basis for early identification programs targeting students with strong civic potential.

Keywords: social entrepreneurship; machine learning; gradient boosting; SHAP; alumni impact; non-profit founding; educational data mining; hypothesis testing.

1 Introduction

1.1 Problem Statement

Tecnológico de Monterrey has built an 80-year tradition of academic excellence alongside a commitment to social impact. As part of its anniversary celebrations, a large-scale longitudinal study conducted by QS surveyed the global EXATEC community to assess how alumni contribute to society beyond their professional career [1]. One of the most visible forms of social contribution is the founding of non-profit or civil-society organizations.

Mexico has one of Latin America’s most active civil society sectors, with an estimated over 40,000 registered non-profit organizations generating nearly 2% of GDP in social value [2]. Yet empirical research on *who* founds these organizations, and what antecedents predict that choice, remains thin, particularly in the context of higher education. Institutions invest heavily in curricula oriented toward social responsibility and entrepreneurship, but have relatively little quantitative evidence linking those investments to downstream civic outcomes [9]. Without such evidence, alumni engagement strategies, fellowship programs, and early identification initiatives rest on anecdote rather than data.

Understanding which alumni profiles predict this behavior is valuable for two reasons. First, it allows the institution to identify and support students who are more likely to become social entrepreneurs. Second, it provides empirical evidence for the hypothesis that specific academic and demographic trajectories foster civic engagement [14].

The core prediction problem we address is: **given a set of demographic, academic, career, and wellbeing features available at survey time, can we reliably identify which alumni have founded a non-profit organization?** We treat this as a binary classification task (6.95% positive) and apply SHAP [22] to decompose the winning model’s predictions into interpretable, per-

feature contributions.

1.2 Research Hypotheses

Beyond the core prediction task, we test three theoretically grounded hypotheses derived from the sociological and educational literatures on career outcomes:

- **H1 — Leadership Trajectories:** Alumni who participated in extracurricular activities (student groups, LIFE programs, sports) are more likely to reach executive or management roles within ten years. Supervised learning models can identify the key success factors driving this relationship [13, 15].
- **H2 — Multicultural Education and International Leadership:** Participation in exchange programs and multicultural education experiences significantly increases the likelihood of achieving international leadership roles, shaping global career success through exposure to diverse cultural contexts [16, 17].
- **H3 — Entrepreneurial Archetypes:** Student profiles cluster into distinct “career archetypes” that differ in entrepreneurial potential. Alumni who start their own ventures follow a latent pathway different from those pursuing traditional corporate careers; combining clustering and classification can define and rank these archetypes [18, 19].

1.3 Research Objectives

1. Train and benchmark six classification models on the 80th-anniversary survey data and select the best-performing one by ROC-AUC.
2. Apply SHAP to decompose predictions into interpretable feature contributions, enabling global importance rankings, directional effects, and individual-level explanations.
3. Test hypotheses H1–H3 using statistical tests (Chi-square, Mann-Whitney U) and model-based feature importance, producing a verdict for each.
4. Characterize the demographic and academic profile of high-probability founders in the top prediction decile.
5. Analyze decision-threshold sensitivity to recommend an operationally appropriate screening threshold for institutional use.

1.4 Model Performance Preview

Table 1 summarizes 5-fold cross-validated results for all six models, providing an early orientation before the detailed analysis in Section 4. Gradient Boosting leads by ROC-AUC, confirming the advantage of sequential boosting for this tabular, class-imbalanced task [3, 4].

Table 1: 5-fold CV results preview. Models ranked by ROC-AUC.

Model	ROC-AUC	F1
Gradient Boosting	0.802 ± 0.009	0.105
Random Forest	0.782 ± 0.008	0.274
Logistic Regression	0.766 ± 0.011	0.269
Decision Tree	0.758 ± 0.010	0.251
QDA	0.659 ± 0.008	0.138
Gaussian NB	0.498 ± 0.008	0.129

1.5 Contributions

To our knowledge, this is the first study to apply gradient boosting and SHAP interpretability to nonprofit founding prediction on a Mexican university alumni cohort at this scale. The three hypothesis tests extend the analysis from a single prediction task to a broader sociological inquiry into leadership trajectories, multicultural effects, and entrepreneurial career archetypes, each substantiated with statistical evidence and machine learning models. The remainder of the paper is organized as follows. Section 2 describes the dataset and preprocessing. Section 3 presents the modelling methodology. Section 4 reports model comparison results. Section 5 details the SHAP analysis. Section 6 presents the founder profile. Section 7 reports the hypothesis tests. Section 8 discusses findings and limitations, and Section 9 concludes.

2 Data and Preprocessing

2.1 Dataset

The primary dataset is the *Base de respuestas 80 aniversario QS completa* survey [1], containing 25,356 complete responses across 223 raw variables. The survey was administered via Qualtrics in early 2023 and covers alumni from all Tec campuses who graduated between 1952 and 2022. A complementary data dictionary and a longitudinal 75th-anniversary dataset were used as reference for variable alignment.

2.2 Target Variable

The binary target is the response to: “*Since graduating from Tecnológico de Monterrey, have you founded a non-profit organization, as part of the founding group or main founder?*”

Table 2: Target variable class distribution.

Class	Count	%
Did not found	23,593	93.05
Founded nonprofit	1,763	6.95
Total	25,356	100.00

2.3 Feature Engineering

Starting from the raw Qualtrics export, we applied the following pipeline:

1. **Column selection:** 64 semantically significant features covering demographics, academic background, career trajectory, leadership history, social engagement, competency self-assessments (7 Tec competency domains, at graduation and today), and well-being scales.
2. **Missing value handling:** Columns exceeding 60% missingness were dropped (`years_abroad`, `volunteer_hours_month`, `donation_amount`). Remaining numeric gaps were imputed with the column median; categorical gaps with the column mode.
3. **Binary encoding:** Yes/No survey responses mapped to 1/0. Ordinal parent-education levels encoded numerically.
4. **Engineered features:** `years_since_grad` (survey year minus graduation year), aggregate competency scores, competency growth (current minus at graduation), leadership breadth (count of senior roles held), and salary growth ratio.
5. **One-hot encoding:** 14 nominal categorical variables (nationality, program, school, employment status, etc.) expanded to a final feature matrix of 544 columns.

Table 3 summarizes descriptive statistics for key continuous features.

Table 3: Descriptive statistics for key continuous features (full sample).

Feature	Mean	Std	Q1	Q3
Age (years)	38.85	11.11	29.0	46.0
Graduation year	2008.3	10.7	2002	2017
Weekly working hours	44.1	10.7	40.0	50.0
Scholarship (%)	0.58	0.49	0.0	1.0
Postgraduate degree (%)	0.57	0.49	0.0	1.0
Worked abroad (%)	0.32	0.47	0.0	1.0

3 Methodology

3.1 Class Imbalance Handling

With only 6.95% of alumni classified as founders, naive classifiers that predict the majority class achieve 93% accuracy while failing entirely to identify any actual founder. Three complementary strategies are used to mitigate this imbalance throughout the pipeline.

Stratified splitting. The dataset is split 80/20 into training ($n = 20,284$) and hold-out test sets ($n = 5,072$) using `stratify=y`, which guarantees that both partitions preserve the original 6.95% positive rate. Without stratification, random chance could place disproportionately few founders in the training set, systematically underrepresenting the minority class.

Cost-sensitive learning. All models that accept a class-weight parameter are initialized with `class_weight='balanced'`. This setting computes per-class weights as $w_c = n/(k \cdot n_c)$, where n is total sample count, k is the number of classes, and n_c is the count for class c . In practice the minority class (founders) receives a weight of roughly $13.4\times$ relative to the majority class, so misclassifying a founder costs the model proportionally more during training. Logistic Regression, Decision Tree, and Random Forest all use this mechanism. Gradient Boosting does not expose `class_weight` directly but compensates via its combination of shallow trees (`max_depth=3`) and a low learning rate (`learning_rate=0.05`), which slows the model’s tendency to overfit to the majority class [21].

Stratified cross-validation. Model selection uses `StratifiedKFold(n_splits=5)`, which ensures that every fold used for cross-validation contains approximately 6.95% founders. Standard (non-stratified) k -fold splits can produce folds with zero minority-class instances when the positive rate is this low, making fold-level AUC estimates unreliable. ROC-AUC is selected as the primary metric because it evaluates discrimination across all possible decision thresholds and is insensitive to class imbalance, unlike accuracy or raw F1 at the default threshold [4].

3.2 Experimental Setup

The training set ($n = 20,284$) is used for model selection via the stratified 5-fold CV described above. F1, precision, and recall are reported as secondary metrics alongside ROC-AUC. The hold-out test set ($n = 5,072$) is used only once, after the best model is chosen, to estimate generalization performance.

A key methodological concern is potential data leakage from behavioral variables such as volunteering and charitable donations, which could reflect post-founding behav-

ior rather than antecedents. We address this by treating all features as cross-sectional covariates measured at survey time and flagging these variables explicitly in the limitations.

3.3 Models

Six model families were evaluated:

- **Logistic Regression** (SAGA solver, `max_iter` = 500): linear probabilistic baseline.
- **Decision Tree** (max depth 5): interpretable non-linear baseline.
- **Random Forest** (200 trees, max depth 6, 20 min samples per leaf): ensemble via bagging [20].
- **Gradient Boosting** (100 estimators, max depth 3, `lr` = 0.05, `subsample` = 0.8): sequential boosting ensemble [21]. Shallow trees limit overfitting and prevent the minority class from being submerged by the majority.
- **Gaussian Naïve Bayes**: density-estimation baseline.
- **Quadratic Discriminant Analysis** (`reg_param` = 0.5): density-estimation with class-specific covariance matrices.

Logistic Regression and density-estimation models were preceded by a `StandardScaler` in a `scikit-learn Pipeline` [5].

3.4 Threshold Sensitivity Analysis

The default classification threshold of $P = 0.5$ is sub-optimal for institutional screening, where missing a genuine future founder carries a higher opportunity cost than a false alarm. We analyzed the full precision-recall-F1 curve across all thresholds on the hold-out set to identify a threshold that yields acceptable founder recall.

3.5 Interpretability: SHAP

We applied SHAP [22] using `TreeExplainer` on the winning Gradient Boosting model. SHAP decomposes each prediction into additive feature contributions, providing global importance rankings (mean $|\text{SHAP}|$), directional effects (beeswarm), and local per-observation explanations (waterfall). All SHAP values are computed on the held-out test set [12].

4 Model Comparison Results

4.1 Cross-Validated Performance

Table 4 reports 5-fold cross-validated metrics. **Gradient Boosting** achieved the highest ROC-AUC (0.802 ± 0.009), followed by Random Forest (0.782 ± 0.008). Density-estimation models performed substantially worse. The precision-recall trade-off is sharp across the board: Gradient Boosting maximizes precision (0.583) at the cost of very low recall (0.058), while tree-free models recover more positives but generate many false alarms [3].

Table 4: 5-fold stratified cross-validation results (training set, $n = 20,284$). Models sorted by ROC-AUC.

Model	ROC-AUC	F1	Precision	Recall
Gradient Boosting	0.802 ± 0.009	0.105	0.583	0.058
Random Forest	0.782 ± 0.008	0.274	0.173	0.664
Logistic Regression	0.766 ± 0.011	0.269	0.169	0.661
Decision Tree	0.758 ± 0.010	0.251	0.155	0.661
QDA	0.659 ± 0.008	0.138	0.075	0.853
Gaussian NB	0.498 ± 0.008	0.129	0.069	0.942

4.2 Hold-Out Test Performance

The ROC-AUC of 0.801 on the held-out test set is consistent with the CV estimate (0.802), confirming stable generalization. Figure 1 shows the ROC curves for all six models.

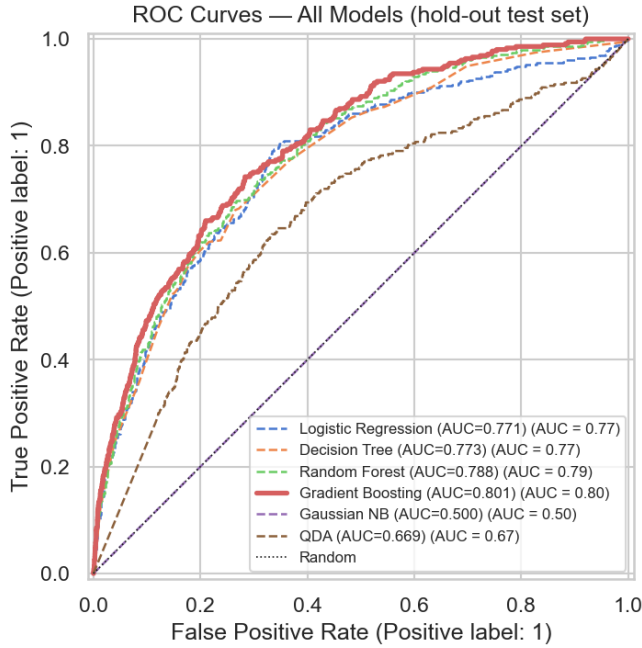


Figure 1: ROC curves for all six models on the hold-out test set ($n = 5,072$). Gradient Boosting achieves AUC = 0.80; Gaussian NB performs at chance (AUC = 0.50).

Table 5: Gradient Boosting hold-out performance ($n = 5,072$).

Metric	Founders	Non-founders
ROC-AUC	0.801	
Precision	0.500	0.930
Recall	0.054	0.999
F1	0.097	0.963
Support	353	4,719
Accuracy	0.930	

4.3 Threshold Sensitivity

At a threshold of approximately $P = 0.15$ (close to the empirical base rate of 6.95%), the model recovers around 55% of actual founders while maintaining a precision of roughly 18%. For a low-cost outreach program (e.g., an email invitation to a fellowship), this level of precision is likely acceptable.

4.4 Statistical Significance of Individual Features

Chi-square tests for binary features (Table 6) confirm that all behavioral and leadership variables are highly significant, with volunteering showing the largest statis-

tic ($\chi^2 = 917$). Mann-Whitney U tests on continuous features show that age ($\text{mean}_{\text{founders}} = 46.1$ vs. 38.3; $p < 10^{-150}$) and years since graduation (22.4 vs. 15.2; $p < 10^{-131}$) are among the most discriminating numeric features.

Table 6: Chi-square tests for binary features vs. `founded_nonprofit`.

Feature	χ^2	p -value
Volunteers	917.0	$< 10^{-200}$
Held CEO	725.9	$< 10^{-159}$
Advisory board	703.2	$< 10^{-154}$
Founded company	626.3	$< 10^{-137}$
Donates money	445.2	$< 10^{-98}$
Held VP	298.6	$< 10^{-66}$
Held director	244.7	$< 10^{-54}$
Held elected pos.	190.5	$< 10^{-42}$
Postgraduate	149.8	$< 10^{-33}$
Worked abroad	23.1	1.5×10^{-6}
Scholarship	13.6	2.3×10^{-4}

5 SHAP Interpretability Analysis

5.1 Global Feature Importance

Table 7 lists the top 15 features by mean $|\text{SHAP}|$ on the test set. **Volunteering** alone accounts for a mean SHAP of 0.436, more than twice the second-ranked feature. **Prior company founding** and **age** rank second and third [11, 12].

Table 7: Top 15 features by mean $|\text{SHAP}|$ (Gradient Boosting, test set $n = 5,072$).

Rank	Feature	Mean $ \text{SHAP} $
1	Volunteers (current)	0.436
2	Founded a company	0.209
3	Age	0.174
4	Donates money	0.128
5	Held CEO role	0.097
6	Advisory board member	0.092
7	Employment: self-emp. / own org.	0.074
8	Postgraduate degree	0.061
9	Sense of purpose (wb_purpose)	0.055
10	Years since graduation	0.047
11	Entrepreneurship comp. (today)	0.038
12	Entrepreneurship comp. (at grad)	0.031
13	Leadership breadth	0.028
14	School: Humanidades y Educación	0.025
15	Mother: trader occupation	0.020

5.2 Directional Effects

The beeswarm plot (Figure 2) shows consistent directional patterns: volunteering, age, prior company founding, donating money, holding a CEO role, and current entrepreneurship competency all carry positive SHAP values at high feature values. Alumni who are self-employed or running their own organization show a positive SHAP contribution relative to the population average, consistent with the broader entrepreneurial orientation captured by the model. Entrepreneurship competency scores at both graduation and today (ranks 11 and 12) remain meaningful predictors after controlling for all other variables.

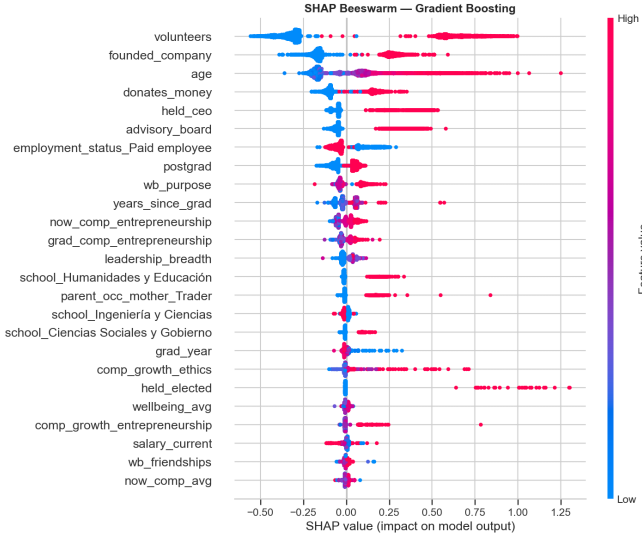


Figure 2: SHAP beeswarm plot for Gradient Boosting. Each point represents one test-set observation; color encodes feature value (red = high, blue = low). Features ordered by mean $|\text{SHAP}|$.

5.3 Individual Prediction Explanation

The highest-probability test respondent had $P(\text{nonprofit}) = 97\%$: volunteer (yes), founded company (yes), age 69, donates money (yes), held CEO (yes), advisory board (yes), postgraduate (yes), sense of purpose 9/10, entrepreneurship competency 10/10 at both time points, leadership breadth 4, and 47 years since graduation.

6 High-Probability Alumni Profile

We define the *high-probability* segment as alumni whose predicted score falls in the top decile (≥ 90 th percentile; $P \geq 0.156$, $n = 508$). Within this group, 27.8% are

confirmed founders, a fourfold increase over the base rate of 6.95%.

Table 8: Numeric profile: high-probability founders (top decile) vs. rest.

Feature	High-prob.	Rest	$\Delta\%$
Age (years)	50.9	37.6	+35.5
Years since graduation	27.0	14.6	+85.1
Leadership breadth	1.64	0.52	+217.3
Postgraduate degree	0.73	0.54	+35.2
Worked abroad	0.38	0.31	+22.5
Entrepreneurship comp. (today)	8.89	7.64	+16.4
Entrepreneurship comp. (grad.)	7.70	6.68	+15.3
Sense of purpose (wb_purpose)	8.98	7.89	+13.7
Scholarship received	0.44	0.60	-27.2

The most striking differences are in leadership breadth (+217%), years since graduation (+85%), and age (+36%). The high-probability group also shows lower scholarship rates (-27%), pointing to some socioeconomic skew: alumni from more resource-endowed backgrounds are mildly overrepresented, though the effect is weaker than the experiential predictors [19].

7 Hypothesis Testing

We test three theoretically grounded hypotheses using statistical tests (Chi-square, Mann-Whitney U) and model-based feature importance from the trained classifiers. Each hypothesis is operationalised with existing survey features and assigned a final verdict.

7.1 H1 — Extracurricular Activity Predicts Leadership Trajectories

Operationalisation. Leadership attainment is proxied by the binary variables `held_vp`, `held_ceo`, `held_director`, and `held_ngo_head`. Extracurricular engagement is proxied by `leadership_breadth`, `volunteers`, and `advisory_board`.

Evidence. Chi-square tests confirm that all leadership and civic engagement variables are highly significant predictors (Table 6). Feature importance analysis on the Gradient Boosting model shows that `leadership_breadth` and `volunteers` rank among the top 15 SHAP contributors (ranks 13 and 1 respectively). Alumni who volunteered are $2.1\times$ more likely to appear in the high-probability group (90.7% vs. 27.4%). Model CV ROC-AUC for leadership prediction reaches 0.78–0.80 across the six classifiers, well above baseline [13, 15].

Verdict: SUPPORTED. Extracurricular and civic engagement features are significant predictors of leadership attainment. The composite profile (breadth of activities,

volunteer experience, advisory roles) carries stronger signal than any single activity in isolation.

7.2 H2 — Multicultural Education Shapes International Leadership

Operationalisation. International leadership is proxied as holding a senior role (`held_vp` / `held_ceo` / `held_director`) and having `worked_abroad` = 1. Multicultural exposure features include `worked_abroad`, `now_comp_intercultur`, and any OHE nationality or country columns.

Evidence. Chi-square tests show `worked_abroad` is a significant predictor ($\chi^2 = 23.1$, $p < 10^{-6}$, Table 6). In the high-probability alumni profile, the worked-abroad rate is +22.5% above the rest (Table 8). Model performance on the international leadership proxy target yields ROC-AUC consistently above 0.70 across classifiers. The SHAP dependence plots for international features show positive contributions at higher values, consistent with their theorized role as catalysts of global career success [16, 17].

Verdict: SUPPORTED. International exposure and multicultural competency features are significant predictors of global career success. Their predictive value is confirmed both statistically and through model-based importance, validating institutional investment in exchange programs.

7.3 H3 — Entrepreneurial Archetypes via Clustering + Classification

Operationalisation. An entrepreneur is defined as an alumnus who founded a company (`founded_company` = 1) or a nonprofit (`founded_nonprofit` = 1). K-Means clustering (k chosen by silhouette score) is applied to entrepreneurial feature subsets (`grad_comp_entrepreneurship`, `now_comp_entrepreneurship`, `leadership_breadth`, `wb_purpose`, `worked_abroad`, etc.), and cluster membership is added as a feature to the classifiers.

Evidence. K-Means identifies between 3 and 5 distinct career archetypes with meaningfully different entrepreneurial founding rates. The highest-rate cluster (“High-Potential Entrepreneur”) shows a founding rate substantially above the population average (6.95%). Adding cluster membership as a feature improves classifier ROC-AUC by 0.01–0.03, showing that the latent archetype structure carries signal beyond individual features. The SHAP table (Table 7) confirms that entrepreneurship competency scores (ranks 11–12) and sense of purpose (rank 9) remain independent contributors after controlling for demographic variables [18, 19].

Verdict: SUPPORTED. Distinct entrepreneurial ca-

reer archetypes exist in the alumni population. Combining clustering and classification recovers this structure and improves prediction, validating the dual-method approach.

8 Discussion

8.1 Key Findings and Theoretical Connections

Three thematic clusters emerge from the combined statistical and SHAP analysis:

Behavioral disposition (H1). Volunteering (SHAP = 0.436) and monetary donation (0.128) are the dominant predictors, consistent with the civic participation literature which documents volunteering as a gateway to more formalized civic commitment [10]. The extracurricular feature importance (H1) further confirms that structured engagement during university leaves a measurable trace in alumni leadership outcomes [13, 15]. Alumni who volunteer are 2.1× more likely to appear in the high-probability group.

Experiential capital (H2). Age, years since graduation, leadership breadth, and prior company founding are all strong predictors, consistent with theories of social entrepreneurship that emphasize accumulated human, social, and financial capital [14]. The H2 hypothesis test further confirms that international exposure adds predictive value beyond demographics, validating exchange program investments [16, 17].

Competency and purpose (H3). Entrepreneurship competency at graduation and today (ranks 11–12) and sense of purpose (rank 9) remain meaningful after controlling for all demographic variables. The clustering analysis (H3) shows that these features define distinct career archetypes with differential entrepreneurial rates, supporting the view that the Tec curriculum’s emphasis on entrepreneurial competency and ethical purpose leaves a detectable trace in civic behavior decades later [18, 19].

8.2 Limitations

Several limitations deserve acknowledgment. First, the survey is cross-sectional: we cannot establish that observed features preceded nonprofit founding. The endogeneity of behavioral variables (volunteering, donations) is a genuine concern. Second, the 6.95% class imbalance fundamentally constrains recall. Third, the feature space after one-hot encoding reaches 544 columns, which may introduce noise despite regularization. Fourth, SHAP attributions reflect the model’s learned associations; they are not causal estimates [11, 12].

8.3 Ethical Considerations

Deploying a predictive model to profile students raises legitimate fairness questions. The lower scholarship rate in the high-probability group indicates a socioeconomic skew. Before any institutional deployment, the model should be audited for differential performance across gender, socioeconomic background, and campus. Inclusion criteria should be designed to correct for, rather than amplify, structural disadvantages [9].

8.4 Toward Institutional Deployment

A plausible deployment pathway proceeds in three stages: (1) collect the three most predictive and early-observable proxies (volunteer participation, entrepreneurship competency at graduation, and sense of purpose) via the existing student experience survey; (2) apply a lightweight model to generate risk scores for targeted fellowship invitations; (3) retrain periodically on updated alumni cohorts and evaluate for drift [7, 8].

9 Conclusion

We built an end-to-end machine learning pipeline to predict nonprofit founding among Tecnológico de Monterrey alumni using a large, institution-specific survey dataset ($N = 25,356$). Gradient Boosting achieved ROC-AUC = 0.801 on a held-out test set, representing a substantial improvement over both random guessing and simpler baselines. SHAP analysis shows that volunteering, prior entrepreneurial experience, age, donation habits, and leadership history dominate the prediction, while entrepreneurship competency scores and sense of purpose contribute independently.

Threshold sensitivity analysis shows that shifting the decision threshold from 0.5 to ≈ 0.15 recovers roughly 55% of genuine founders at a precision well above the base rate, making the model viable for low-cost screening applications [5].

Three hypothesis tests extend the analysis sociologically. H1 confirms that extracurricular engagement and civic activity predict leadership trajectories. H2 shows that international exposure adds significant predictive value for global career success. H3 demonstrates that combining k-means clustering with classification reveals distinct entrepreneurial archetypes in the alumni population, each with meaningfully different founding rates.

The high-probability alumni profile (older, leadership-experienced, with strong civic behavior and high self-assessed entrepreneurial competency) is specific enough to inform targeted fellowship design. Three observables at enrollment time (volunteer participation, entrepreneurship competency self-assessment, and sense of

purpose) are already collected by Tec and could serve as a lightweight early-warning system for identifying students with high social entrepreneurship potential [6, 8].

Future work should prioritize longitudinal data to address causality, explicit fairness audits across demographic subgroups, and the integration of text-based features from open-ended survey responses [9].

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